

## Project Proposal

### IE 7275: Data Mining in Engineering

#### **Problem Setting**

Retail businesses have limited operating and advertising budgets, and so many small businesses have come to operate exclusively online to reduce overhead costs. While raw data can be generated and collected more easily and in larger quantities online, ease of interpretability is reduced compared to a brick-and-mortar storefront where an operator or owner may be able to easily infer which customers are more likely to make a purchase based off visual cues. Without the ability to make inferences from these data effectively, smaller e-tailers risk misallocating resource expenditure towards ineffective campaigns and site designs.

#### **Problem Definition**

The goal of the project is to predict if a visitor will make a purchase based only on their site activity during a given browsing session. Beyond this, secondary goals are to identify elements of the digital storefront that are most effective or ineffective in converting visitors to customers, and behavioural patterns shared by converting customers.

#### **Data Source**

The dataset used is the Online Shoppers Purchasing Intention dataset from the UCI Machine Learning Repository. The data were gathered from a commercial online retail site, with bounce/exit rates and page value captured through Google Analytics.

<https://doi.org/10.24432/C5F88Q>

#### **Data Description**

The data contain 12,330 records of customers browsing an online store, where each record corresponds to a single visit from a unique user obtained during a one-year period. There is a total of 18 columns in the dataset, of which 1 is the binary target response (revenue). Of the other variables, one is binary, two are categorical, three are continuous, and eleven are integers. The dataset is class-imbalanced, with 10,422 entries, or 84.5%, resulting in no purchase, and only 1908 positive responses generating revenue. All entries in the dataset are complete, with no

missing values, but beyond the class imbalance, there are some predictors with imbalanced categories, which may require special consideration.

## **Methodology**

The binary and categorical variables will be one-hot encoded, which will also increase the number of predictors. An exploratory data analysis will be conducted, including visual overviews of the data, to evaluate variable distribution patterns and correlation between predictors. The data will be split 80:20 into training and test sets with a predetermined seed, stratified on the target variable to ensure the same distribution in both sets.

Some of the binary classification models I intend to test currently include binomial logistic regression without regularization, which will serve as an easily interpretable baseline against which to measure our other modelling methodologies. Others include Binomial LASSO, which will help perform feature selection and identify the most significant contributors towards purchase intent amongst all variables in the dataset; Random forest, which works on all variables and will also help to identify important features; and XGBoost, which is known to perform well on datasets with notable class imbalances. These models will require some degree of hyperparameter tuning, in the case of LASSO (which requires tuning regularization strength), random forests (which require specification of the number of trees used as well as the depth per tree), and XGBoost (which accepts arguments for its trees and shrinkage), which can be done in a loop for performing 10-fold cross-validation.

Due to the class imbalance, I intend to randomly oversample the entries with positive responses in the training set prior to training the LASSO and logit models. The tree-based models are more robust to imbalances in data. Simple vote-based ensemble models of the most effective methods will also be tested. PCA will likely not be necessary for this dataset as the number of predictors is still within reason, particularly after feature selection, and PCA would detract from ease of interpretability of each model to a layperson. Accordingly, PCA will only be considered if the initial data exploration suggests strong multicollinearity.

Finally, because of the heavy class imbalance, models will be evaluated on F1 score and confusion matrices rather than on accuracy alone, as a naïve classifier would be able to achieve 84.5% accuracy simply by predicting that no site visits will result in a sale.